ANSWER KEY & MARKING SCHEME SET 1

Chapter 2: **Exploring Mixtures and Their Separation** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 20 per cent	30/150.
2	**(a)** fractional distillation	Boiling-point difference.
3	**(d)** sublimation, dissolving and filtering, evaporation	Camphor first.
4	**(a)** 10 g	4/100 x 250.
5	**(d)** sublimation	Iodine sublimes.
6	**(c)** is collected first	Evaporates first.
7	**(a)** 50 mL	0.2 x 250.
8	**(a)** separating funnel	Immiscible.
9	**(c)** colloidal particles	Particles smaller than pores.
10	**(b)** 470 g	Solute 30 g.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	8/100 x 500 (1); = 40 g (1).	1+1
12	To measure the temperature of the vapour (1); Which equals the boiling point of the liquid (1).	1+1
13	Dissolved salts raise the boiling point (1); Solute particles reduce vapour pressure (1).	1+1
14	Solution = 100 g (1); 12 per cent (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	36/200 x 100 = 18 per cent (1); New solution = 100 g (1); New percentage = 36 per cent (1).	1+1+1
16	KNO3 (1); Its solubility falls sharply on cooling (1); NaCl solubility changes little (1).	1+1+1
17	Pure liquid has a fixed boiling point (1); Mixture has components with different b.p. (1); Composition changes (1).	1+1+1
	OR: Acetic acid = 25 mL (1); Add to some water (1); Make up to 250 mL (1).	1+1+1
18	Size (1); Tyndall effect (1); Examples: sugar solution, milk (1).	1+1+1
	OR: Soap and milk are colloids and scatter (1); Copper sulphate is a true solution (1); No path visible in it (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	A = 5 per cent (1); B = 10 per cent (1); C = 15 per cent (1); Mixing equal masses of A and C gives 10 per cent (1); Ranking C > B > A (1).	1+1+1+1+1
	OR: Air filtered, CO2 and water removed (1); Compressed and cooled (1); Liquid air fractionated (1); Nitrogen, argon, oxygen collected (1); Uses of each gas (1).	1+1+1+1+1
20	Magnet: iron (1); Sublimation: ammonium chloride (1); Dissolve and filter: sand (1); Evaporate: salt (1); Principles: magnetic, sublimation, solubility, boiling point (1).	1+1+1+1+1
	OR: Crush leaves in alcohol (1); Spot on filter paper (1); Dip in solvent (1); Several coloured bands form (1); Chlorophyll, carotene, xanthophyll (1).	1+1+1+1+1

Mistakes to avoid:

- Calculating mass per cent using the mass of the *solvent* instead of the mass of the *solution* (solute + solvent).
- Saying that a clear liquid must be a pure substance; a clear solution such as salt water is a mixture.
- Using filtration to separate a dissolved solid; filtration works only for insoluble particles, not for a solute.
- Confusing evaporation (solvent lost, solid left) with distillation (solvent collected by condensing); and using fractional distillation when the boiling points differ by more than 25 K.
- Thinking that the solubility of a gas also increases when the temperature rises; the solubility of gases in water decreases.
- Forgetting that a colloid does not settle and cannot be filtered, but still shows the Tyndall effect.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 2: **Exploring Mixtures and Their Separation** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** crystallisation	Cool saturated solution.
2	**(d)** colloid or suspension	Scattering.
3	**(c)** 13.3 per cent	20/150.
4	**(b)** R_f value	Chromatography.
5	**(d)** crystals separate	Solubility decreases.
6	**(a)** 10 per cent	20/200.
7	**(b)** 0.5	4/8.
8	**(a)** winnowing	Wind blows lighter husk.
9	**(a)** 100 g	2.5 x 40.
10	**(c)** magnetic separation	Magnetic property.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Shine a beam of light (1); Path is visible due to scattering (1).	1+1
12	No fixed composition (1); Components keep properties of metals (1).	1+1
13	R_f = 3.2/8.0 (1); = 0.4 (1).	1+1
14	They are immiscible (1); Oil is less dense and floats (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Magnet removes iron (1); Dissolve and filter sand (1); Evaporate for salt (1).	1+1+1
16	Spin at high speed (1); Denser cells move to the bottom (1); Plasma stays above (1).	1+1+1
17	Add KNO3 in 100 g of water at 30 and at 60 degrees Celsius (1); Find the mass dissolved (1); Larger at higher temperature (1).	1+1+1
	OR: Round bottom flask, thermometer (1); Condenser with water flow (1); Receiver for the distillate (1).	1+1+1
18	Immiscible (1); Oil has lower density (1); Oil floats above (1).	1+1+1
	OR: (a) Sublimation (1); (b) Fractional distillation (1); (c) Chromatography (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Heated and fed to column (1); Petrol, kerosene (1); Diesel, lubricating oil (1); Bitumen at the bottom (1); Different b.p. ranges (1).	1+1+1+1+1
	OR: Pure substance has a sharp melting point (1); Pure liquid has a fixed boiling point (1); Mixtures melt over a range (1); Boiling point rises as mixture boils (1); Compare with known values (1).	1+1+1+1+1
20	Distillation setup (1); Condenser (1); Collect at 100 degrees Celsius (1); Test with silver nitrate for chloride (1); Residue is salt (1).	1+1+1+1+1
	OR: Screening (1); Sedimentation with alum (1); Filtration (1); Chlorination (1); Storage and distribution (1).	1+1+1+1+1

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- Thinking that the solubility of a gas also increases when the temperature rises; the solubility of gases in water decreases.
- Forgetting that a colloid does not settle and cannot be filtered, but still shows the Tyndall effect.

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MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 2: **Exploring Mixtures and Their Separation** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** fractional crystallisation	Solubilities vary with temperature.
2	**(c)** 40 g	40 g in 100 g water.
3	**(b)** milk, fog, butter	Dispersed particles.
4	**(b)** fractional distillation	Close b.p.
5	**(a)** temperature increases	Gas escapes on warming.
6	**(a)** has a fixed melting point	Sharp melting point.
7	**(d)** it is less dense	Lower density.
8	**(a)** milk	Colloid.
9	**(a)** electrolysis of water	Chemical decomposition.
10	**(b)** colloid (solid in gas)	Aerosol.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Mass of solute is the same (1); Percentage decreases (1).	1+1
12	Solubility decreases at lower temperature (1); Excess solute separates as crystals (1).	1+1
13	To fit tightly in the funnel (1); Avoid air bubbles and speed filtration (1).	1+1
14	Particles are smaller than filter pores (1); They pass through (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Column provides many condensation steps (1); Better separation (1); Pure fractions obtained (1).	1+1+1
16	Dissolve in water (1); Filter (1); Evaporate and crystallise (1).	1+1+1
17	Components melt at different temperatures (1); Impure substances melt over a range (1); Composition varies (1).	1+1+1
	OR: More soluble moves faster (1); Strongly held stays near the base (1); Bands appear at different heights (1).	1+1+1
18	Liquefaction and fractional distillation (1); Absorption in a suitable liquid (1); Diffusion through membrane (1).	1+1+1
	OR: Sea water is let into shallow pans (1); Sun evaporates water (1); Salt crystals are collected and purified (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Add salt in portions at 30 degrees Celsius (1); Stop when some remains (1); Prepare a hot saturated solution (1); Cool to get crystals (1); Filter and dry (1).	1+1+1+1+1
	OR: Fractional distillation (1); Ethanol b.p. 78, water 100 (1); Ethanol collects first (1); Some water also vaporises (1); Repeat or use a longer column (1).	1+1+1+1+1
20	Particle size (1); Appearance (1); Tyndall effect (1); Stability (1); Examples (1).	1+1+1+1+1
	OR: Miscible: alcohol-water (1); Immiscible: oil-water (1); Solute and solvent: sugar and water (1); Solubility defined (1); Solution defined (1).	1+1+1+1+1

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